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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395
Applied Cyber Security

Midterm Examination (MTA) - Spring Semester 2025/2026

Date: 16 /APR./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks:**30**
- This exam consists of **3** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1 :Choose The correct answer [10 marks]::

1. Which term refers to protecting information from unauthorized access?
   A. Availability
   B. Confidentiality
   C. Integrity
   D. Authentication
2. Malware that disguises itself as legitimate software is called:
   A. Worm
   B. Trojan
   C. Spyware
   D. Rootkit
3. Which attack relies on manipulating human behavior rather than technology?
   A. DDoS
   B. Phishing
   C. SQL Injection
   D. Buffer overflow
4. A virus differs from a worm because a virus:
   A. Spreads automatically
   B. Requires user action
   C. Attacks networks only
   D. Is hardware-based
5. Which attack floods a system with traffic to make it unavailable?
   A. Spoofing
   B. Sniffing
   C. DoS
   D. Man-in-the-middle
6. Which system detects attacks but does not block them?
   A. IPS
   B. Firewall
   C. IDS
   D. VPN
7. A VPN mainly provides:
   A. Authentication only
   B. Encryption over public networks
   C. Malware detection
   D. Load balancing
8. Which wireless security protocol is the most secure?
   A. WEP
   B. WPA
   C. WPA2/WPA3
   D. Open authentication
9. Which control is an example of a preventive control?
   A. Log review
   B. Firewall
   C. Audit report
   D. Incident response
10. Phishing is a type of:
    A) Malware
    B) Social engineering attack
    C) Firewall attack
    D) Encryption method

Question 2: [10 Marks]

1. How does a worm propagate within a network environment? Explain how malware can affect system performance and security? (3 mark)
2. Explain and compare Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) in terms of their functions and roles in network security. (3 mark)
3. How does a firewall contribute to network protection? (2 mark)
4. How does phishing manipulate human psychology to achieve its goal? (2 mark)

Question 3: [10 marks]:

Case Study (1); A university deployed wireless access points using WEP encryption to simplify student connectivity. The SSID was hidden, and MAC filtering was enabled.

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